

KILLARA HIGH SCHOOL



YEAR 11X1-MATHEMATICS HSC Assessment 1- Term 4, 2007

TOTAL TIME: 70 minutes

TOTAL MARKS: 40

Read each question thoroughly.

Attempt all questions.

Start each question on a new page.

Non-Programmable Calculator may be used.

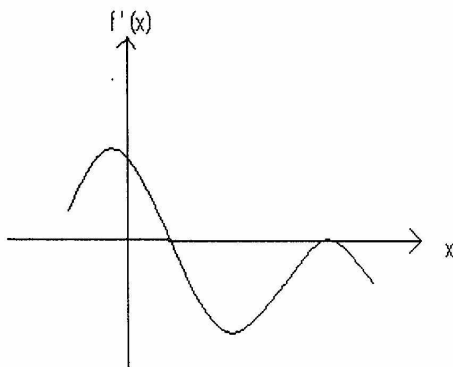
All necessary working out must be shown to obtain full marks.

Question 1**marks**

- (a) Evaluate $\sum_{n=3}^{20} 4n - 2$ **2**
- (b) The number 2, a, b are in arithmetic progression and a, b, 9 are in geometric progression. Find the values of a and b. **3**
- (c) Pat borrows \$45000 at 7.2% per annum reducible interest, calculated monthly. The loan is to be repaid in 70 equal monthly instalments. Let M be the value of each instalment.
- i. Show that the amount (A) owing after the 70th installment is **3**
$$A_{70} = 45000(1.006)^{70} - \frac{M(1.006^{70} - 1)}{.006}.$$
 - ii. Show that the monthly installment is \$789.18. **1**
 - iii. Find the amount owing after the 11th payment. **1**
 - iv. With the twelfth payment, Pat pays an additional \$8000, so this payment is \$8789.18. After this, repayments continue at \$789.18 per month. How many more repayments will be needed? **3**

Question 2**Marks**

- (a). The following diagram shows the graph of $f'(x)$. Copy the diagram into your writing booklet and sketch a possible graph of $f(x)$. Label clearly any stationary points and points of inflexion. **4**



- (b) Given that $\frac{d^2y}{dx^2} = x - 2$. The point $(0, 3)$ is on the curve and the tangent at this point is parallel to the line $y - 2x + 5 = 0$. Find the equation of the curve. **2**
- (c) A closed cylindrical can is to be made out of 294π square centimeters of sheet metal. If h is the height and r is the radius show that
- $h = \frac{147}{r} - r$. **1**
 - Hence find an expression for the volume in term of r **1**
 - Find the greatest possible volume of this can (answer to the nearest whole number). **3**
- (d) If $y = (2 - x)\sqrt{x}$
- State the domain of the function and find the axes intercepts. **2**
 - Show that $\frac{dy}{dx}$ is undefined at $x = 0$ and that the stationary point occurs at $x = \frac{2}{3}$. Determine its nature. **3**
 - Use these results to sketch the curve $y = (2 - x)\sqrt{x}$. **2**

HSC Assessment 1

Question 1 (13 marks)

(a) 10, 14, 18

$a = 10, d = 4, l = 78$ ①

$S_{18} = \frac{18}{2}(10 + 78)$
 $= 792$ ①

(b) $b - a = a - 2$

$2a = b + 2$

$\frac{b}{a} = \frac{a}{b}$

$b^2 = qa \quad \text{sub. } 2a$

$\Rightarrow b^2 = \frac{q}{2}(b + 2)$ ①

$2b^2 - qb - 18 = 0$
 $(2b + 3)(b - 6) = 0$

or $b = 6$ ①

$\therefore \begin{cases} b = -\frac{3}{2} \\ a = \frac{1}{4} \end{cases}$ or $a = 4$ ①

i) $A_1 = 45000(1.006) - M$ ①

$A_2 = 45000(1.006)^2 - M(1.006) - M$

⋮

$A_{70} = 45000(1.006)^{70} - M(1.006^{69} + 1.006^{68} + \dots + 1)$ ①

$= 45000(1.006)^{70} - \frac{M(1.006^{70} - 1)}{0.006}$ ①

ii) After 70th payment, $A_{70} = 0$

$\therefore 45000(1.006)^{70} - \frac{M(1.006^{70} - 1)}{0.006} = 0$

$M = \frac{45000(1.006)^{70} \times 0.006}{1.006^{70} - 1} = \789.18 ①

iii) $A_{11} = 45000 \times 1.006^{11} - \frac{789.18(1.006^{11} - 1)}{0.006} = \39114.57 ①

iv) After 12th payment, the amount owing is:

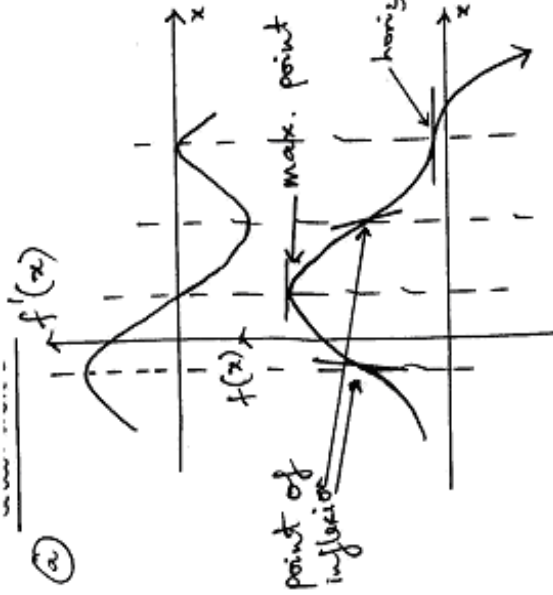
$39114.57 \times 1.006 - 789.18 = \$30,560.58$

$A_n = 30560.58 \times 1.006^n - \frac{789.18(1.006^n - 1)}{0.006} = 0$ ①

$\therefore 1.006^n (30560.58 - \frac{789.18}{0.006}) = -\frac{789.18}{0.006}$ ①

$-100969.42 \times 1.006^n = -131530$ ①

$1.006^n = 1.30267 \Rightarrow n = 44.2 \text{ months.}$ ①



(b) $\frac{dy}{dx} = \frac{x^2}{2} - 2x + c$

$\frac{dy}{dx} = 2$ at $x = 0$

$\therefore x^2 - 2x + c_1 = 2$ at $x = 0$

$\Rightarrow c_1 = 2$

$\frac{dy}{dx} = \frac{x^2}{2} - 2x + 2$

$y = \frac{1}{6}x^3 - x^2 + 2x + c_2$ Sub (0,3)

$\Rightarrow 3 = c_2$

hence the equation of the curve

$y = \frac{1}{6}x^3 - x^2 - 2x + 3$

Marking for (a)

- ① - correct intervals where the curve is increasing & decr.
- ① - horizontal pt. of inflection
- ① - points of inflection
- ① - maximum point and correct shape

① for each error

(i) $294\pi = 2\pi r^2 + 2\pi r h$
 $h = \frac{294\pi - 2\pi r^2}{2\pi r}$

$= \frac{147}{r} - r$

(ii) $V = \pi r^2 h$

$= \pi r^2 \left(\frac{147}{r} - r \right)$

$= 147\pi r - \pi r^3$

(iii) $\frac{dV}{dr} = 147\pi - 3\pi r^2$

$147\pi - 3\pi r^2 = 0$

$r^2 = 49$

$\therefore r = 7$

hence $h = \frac{147}{7} - 7 = 14$

$\therefore V = \pi \times 7^2 \times 14 = 2155 \text{ cm}^3$

$f''(r) = -6\pi r$

$f''(r) < 0$

$\therefore r=7$ is a max.

① if not showing $f''(r) < 0$

(a) i) $\frac{4}{3}x^3 + \frac{6}{\sqrt{x}} + 2x + C$ (2)

ii) $-\frac{2}{9}(5-3x)^{3/2} + C$ (1)

iii) $\frac{-3}{8(4x^2-1)^2} + C$

(b) $\int_2^5 \frac{3x^4 - 5x^3}{x^2} dx, \int_2^5 3x^2 - 5x dx$

$= \left[x^3 - \frac{5}{2}x^2 \right]_2^5$ (1)

$= (125 - \frac{125}{2}) - (8 - \frac{10}{2}) = 59.5$ (1)

(c)

x	0	0.5	1	1.5	2
$f(x)$	$\frac{1}{4}$	$\frac{1}{2353}$	$\frac{1}{5}$	$\frac{1}{16}$	$\frac{1}{8}$

(1)

$n = 4$
 $h = \frac{2-0}{4} = 0.5$

$A \approx \frac{0.5}{3} \left(\frac{1}{4} + \frac{1}{5} + 4 \times \frac{1}{2353} \right) + \frac{0.5}{3} \left(\frac{1}{5} + \frac{1}{8} + 4 \times \frac{1}{16} \right)$ (1)
 ≈ 0.39 Sq. units. (1)

d) $y = (2-x)\sqrt{x}$

i) domain: $x \geq 0$ (1)

when $x=0, y=0$

when $y=0, x=2$ (1)

ii) $\frac{dy}{dx} = -1\sqrt{x} + \frac{2-x}{2\sqrt{x}}$ (1)

$\therefore \frac{dy}{dx}$ is undefined at $x=0$

for stationary point $\frac{dy}{dx} = 0$

$\Rightarrow -\sqrt{x} + \frac{2-x}{2\sqrt{x}} = 0$

$-2x + 2 - x = 0$

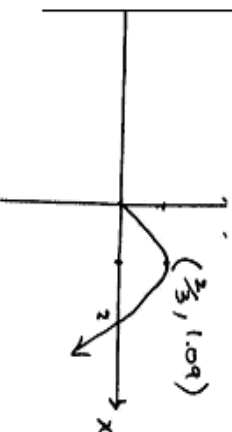
$-3x = -2$

$x = \frac{2}{3}$ (1)

$y = 1.09$

x	$\frac{1}{3}$	$\frac{2}{3}$	1
$\frac{dy}{dx}$	x	0	x
$\frac{dy}{dx}$		$\rightarrow$	

$(\frac{2}{3}, 1.09)$ is a maximum (1)



(1) - Labelling

(1) - Correct shape